EUROSTAT · 2025 EDITION

European Cost of Living Analysis

A detailed project report on income, price levels and effective purchasing power across the 27 EU member states — combining a Python analytics pipeline, statistical modelling, a Power BI dashboard, and an interactive Streamlit application.

Scope	27 EU member states, 2025
Primary sources	Eurostat earn_nt_net · Eurostat prc_ppp_ind_1
Methods	Linear regression · Pearson correlation · residual analysis · affordability index
Delivery	Python (pandas, NumPy, matplotlib) · Power BI · Streamlit

1. Executive Summary

This project investigates the relationship between **income levels, consumer price levels, and effective purchasing power across the 27 European Union member states in 2025**.

The analysis combines two Eurostat datasets:

- **Annual net earnings** for a single person without children earning 100% of the national average wage.
- **Price Level Indices (PLI)** for household final consumption expenditure, with the EU27 average normalized to 100.

The core objective was not simply to identify which countries have the highest salaries or lowest prices, but to examine the more meaningful question:

> *How does the purchasing power implied by national earnings change once differences in price levels are taken into account?*

The project developed a reusable Python data-analysis pipeline for importing, validating, cleaning, transforming and analysing the datasets. Statistical analysis included linear regression, Pearson correlation, residual analysis and derived affordability measures. The results were then exposed through both a **Power BI dashboard** and an interactive **Streamlit application**.

The results show a very strong positive relationship between price levels and nominal net earnings. For the 100%-average-earner scenario, the analysis produced a **Pearson correlation of 0.937** and an **R^2 of 0.877**. In other words, countries with higher consumer price levels generally also report substantially higher nominal net earnings.

However, the regression also reveals important deviations from this general pattern. Some countries have substantially higher or lower earnings than would be expected from their price level alone. This is where the affordability analysis becomes more informative than a simple salary ranking.

Using the project's affordability index, where the EU27 average is set to 100:

- **Luxembourg** had the highest affordability index at approximately **153.2**.
- **Austria** followed at approximately **120.9**.
- **Ireland** was approximately **120.7**.
- **Netherlands** was approximately **115.1**.
- **Hungary** had the lowest index at approximately **62.1**.
- **Greece** followed at approximately **63.9**.
- **Slovakia** was approximately **68.4**.
- **Estonia** was approximately **73.6**.
- **Romania** was approximately **75.5**.

The alternative 50%-earnings scenario produced a lower affordability index across most countries and demonstrated that **income level materially changes the relative purchasing-power picture**, even though the underlying price levels remain unchanged.

The project therefore demonstrates an important distinction between **nominal earnings**, **price levels**, and **effective affordability**.

2. Project Objectives

The project was designed around several analytical objectives.

Primary objective

To compare the economic position of EU countries by combining:

- national net earnings;
- consumer price levels;
- earnings relative to the EU average;
- earnings relative to national price levels.

Secondary objectives

The project also aimed to:

- build a reproducible data-processing workflow;
- validate Eurostat data before analysis;
- remove non-country aggregate observations where appropriate;
- calculate derived purchasing-power indicators;
- quantify the relationship between earnings and prices;
- identify countries performing above or below the expected earnings-price relationship;
- compare different income scenarios;
- communicate the results through interactive visualizations.

The project was therefore intentionally designed as a complete **data analytics workflow**, rather than a single statistical exercise.

3. Research Questions

The analysis addresses the following questions.

RQ1 — Where are consumer prices highest and lowest?

How different are household consumption price levels across EU countries?

RQ2 — Where are nominal net earnings highest and lowest?

Which countries provide the highest and lowest annual net earnings for the selected worker profile?

RQ3 — Are high prices associated with high earnings?

Is there a systematic relationship between a country's price level and its nominal net earnings?

RQ4 — Which countries earn more or less than expected?

Given a country's price level, does its actual net earnings level sit above or below the regression trend?

RQ5 — Which countries offer greater effective affordability?

When earnings are considered together with price levels, which countries appear to provide stronger or weaker purchasing power?

RQ6 — How does the picture change at a lower earnings level?

Does the relative affordability ranking change when the worker earns 50% rather than 100% of the national average wage?

4. Data Sources

The project uses publicly available Eurostat data.

4.1 Annual net earnings

The earnings dataset used was Eurostat's **earn_nt_net — Annual net earnings** dataset. Eurostat describes this as an annual dataset and its current coverage extends through 2025.

The selected earnings definition was:

> *Single person without children earning 100% of the average earnings.*

The analysis used **net annual earnings in euros**.

A second dataset was used for the 50%-earnings scenario.

The project therefore distinguishes between:

- net_earnings_eur
- net_earnings_50_eur

rather than treating “average salary” as a single universal concept.

4.2 Price Level Index

The second major source was Eurostat's **prc_ppp_ind_1** dataset covering Purchasing Power Parities, Price Level Indices and related expenditure measures.

The selected indicator was:

> *Price Level Indices — Household final consumption expenditure.*

The index is normalized so that:

$$EU27 = 100$$

Therefore:

- PLI = 100 → approximately EU-average price level
- PLI > 100 → more expensive than the EU average
- PLI < 100 → less expensive than the EU average

Eurostat's 2025 publication confirms that household consumption price levels varied considerably across the EU, from approximately 63% of the EU average in Bulgaria to approximately 140% in Denmark.

Eurostat also explicitly notes that these price-level comparisons **are not adjusted for differences in income or wage levels**. That distinction is central to this project because the analysis combines PLI with earnings to construct an additional affordability measure.

5. Analytical Scope

The main analysis covers the **27 EU member states**.

The Eurostat files also contain aggregate observations such as:

```
EU27_2020
```

These observations were excluded from country-level calculations where appropriate.

Similarly, countries outside the EU were excluded from the main EU comparison.

This distinction is important because including the EU aggregate in country statistics would artificially alter:

- means;
- standard deviations;
- correlations;
- regression coefficients;
- rankings.

The project therefore separates **country observations** from **reference/aggregate observations**.

6. Data Processing Pipeline

The project was organized into a reusable Python structure rather than keeping all calculations inside a single notebook.

The main components were separated into:

```
src/
■■■ data_processing.py
■■■ analysis.py
■■■ visualization.py
```

This separation provides three conceptual layers.

Data processing

Responsible for:

- loading CSV files;
- previewing datasets;
- checking data quality;
- identifying duplicates;
- checking missing observations;
- excluding unwanted geographic codes;
- retrieving reference values.

Analysis

Responsible for:

- regression;
- correlation;
- earnings indices;
- earnings-to-PLI calculations;
- affordability calculations;
- residual analysis.

Visualization

Responsible for:

- ranked bar charts;
- scatter plots;
- regression lines;
- dumbbell/change charts;
- presentation-oriented plotting.

This architecture makes the analytical logic reusable and reduces dependence on notebook-specific variables.

7. Data Quality Checks

Before calculations were performed, the data was systematically inspected.

The checks included:

- dataset dimensions;
- column names;
- data types;
- missing values;
- duplicate geographic codes;
- selected year;
- selected currency;
- selected earnings case;
- selected earnings structure;
- geographic coverage.

For the main earnings dataset, the analysis found:

- 28 observations initially;
- 27 country observations after excluding the EU aggregate;
- no duplicate country codes;
- no missing OBS_VALUE values.

For the PLI dataset:

- 32 observations initially;
- 27 EU country observations after excluding non-EU countries and the EU aggregate;
- no duplicate country codes;
- no missing OBS_VALUE values.

This validation was particularly important because Eurostat SDMX/CSV exports contain numerous metadata columns, many of which are intentionally empty for the observations being used.

8. Price Level Analysis

The 2025 PLI results show substantial differences in household consumption prices across the EU.

The project's values were approximately:

Country	PLI	Country	PLI
Denmark	139.7	Slovenia	89.3
Ireland	136.2	Czechia	89.4
Luxembourg	131.5	Cyprus	89.2
Finland	120.8	Greece	87.4
Sweden	121.0	Portugal	86.6
Belgium	116.2	Slovakia	85.2
Netherlands	115.6	Lithuania	82.8
Austria	113.0	Latvia	83.2
France	110.3	Croatia	78.4
Germany	108.3	Hungary	77.5
Estonia	101.2	Poland	73.3
Italy	97.1	Romania	65.1
Malta	91.9	Bulgaria	62.5
Spain	91.6		

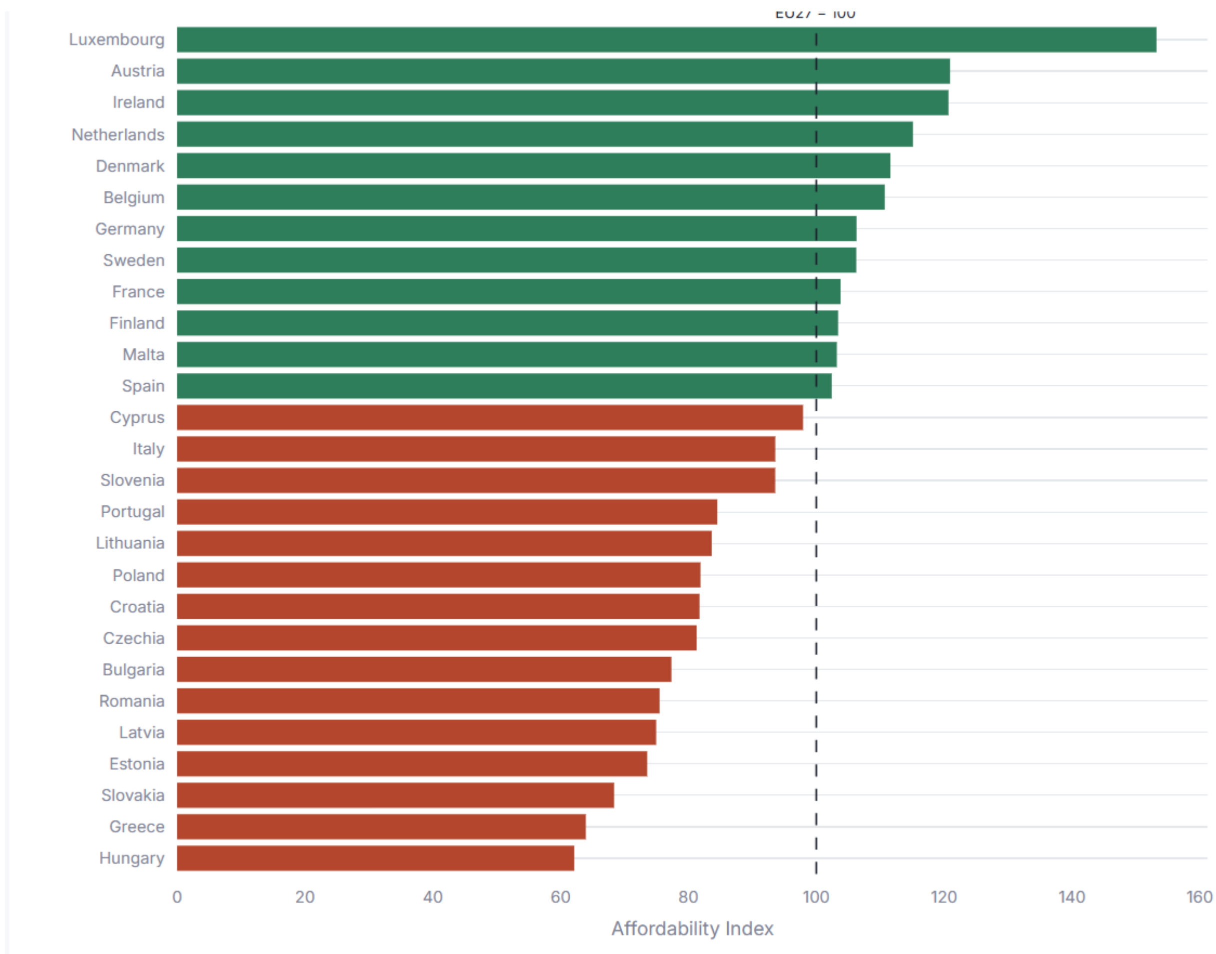

Figure 1. Price Level Index by country, EU27 = 100.

The broad pattern is consistent with Eurostat's published 2025 results. Denmark, Ireland and Luxembourg are among the highest-price EU countries, while Bulgaria, Romania and Poland are among the lowest.

The difference between the extremes is substantial.

Using the project's values:

$$\text{Denmark / Bulgaria} \approx 2.24$$

Therefore, the Danish household-consumption price level is more than twice the Bulgarian level relative to the EU benchmark.

This illustrates why nominal income alone is insufficient for a meaningful cross-country purchasing-power comparison.

9. Net Earnings Analysis

For the 100%-average-earner scenario, the project found an EU27 reference value of:

€26,928.97

The highest observed annual net earnings were:

Country	Net earnings
Luxembourg	€54,259.93
Ireland	€44,263.30
Denmark	€41,981.24
Austria	€36,797.90
Netherlands	€35,836.93
Belgium	€34,642.30
Sweden	€34,624.37
Finland	€33,641.04

At the lower end:

Country	Net earnings
Hungary	€12,967.22
Romania	€13,232.74
Bulgaria	€13,016.65
Greece	€15,049.76
Slovakia	€15,685.92
Poland	€16,163.12

The important observation is that these rankings cannot be interpreted independently from price levels.

For example:

- Luxembourg has exceptionally high earnings **and** exceptionally high prices.
- Romania has relatively low earnings **but also relatively low prices**.
- Denmark has very high earnings **and** the highest PLI.
- Bulgaria has very low earnings **and** the lowest PLI.

This motivates the project's combined affordability analysis.

10. Earnings vs. Price Level Regression

One of the central analyses was a linear regression of:

Net earnings vs. Price Level Index

The model can be represented as:

$$Earnings\_i = \beta\blacksquare + \beta\blacksquare \times PLI\_i + \varepsilon\_i$$

The project calculated the regression using:

```
fit_linear_regression(x, y)
```

The function returns:

- slope;
- intercept;
- predicted earnings;
- residuals;
- Pearson correlation;
- R^2.

For the 100%-earnings scenario:

Metric	Value
Pearson correlation (r)	0.937
Coefficient of determination (R^2)	0.877

This is a very strong positive relationship.

An R^2 of 0.877 means that approximately **87.7% of the variation in nominal net earnings across the analyzed countries is associated with the linear relationship with PLI in this sample**.

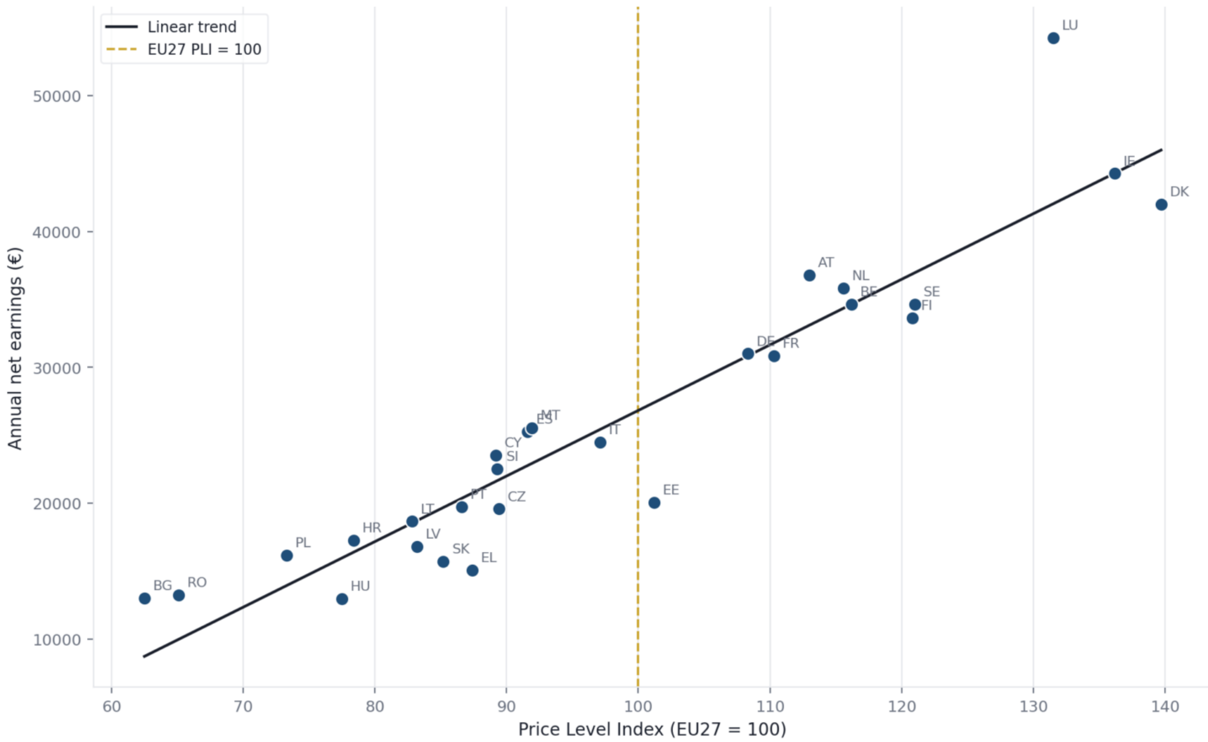

Figure 2. Earnings vs. Price Level, EU27 = 100.

However, this should **not** be interpreted as proof that higher prices cause higher wages.

There are many potential underlying factors, including:

- productivity;
- taxation;
- labor-market institutions;
- economic structure;
- sector composition;
- housing costs;
- national wage-setting systems;
- capital intensity;
- education and skills;
- general economic development.

The regression is therefore best interpreted as a **descriptive cross-country relationship**, not a causal model.

11. Regression Residual Analysis

The regression becomes particularly useful when examining the residuals.

The residual is defined as:

$$Residual\_i = ActualEarnings\_i - PredictedEarnings\_i$$

A positive residual means:

> *Actual earnings are higher than the regression would predict given the country's PLI.*

A negative residual means:

> *Actual earnings are lower than predicted given the country's PLI.*

The largest positive residuals in the project included:

Country	Residual
Luxembourg	+€12,230.92
Bulgaria	+€4,296.97
Austria	+€3,699.65
Romania	+€3,257.93
Malta	+€2,631.60
Spain	+€2,495.21

The largest negative residuals included:

Country	Residual
Estonia	–€7,357.16
Greece	–€5,690.24
Denmark	–€4,006.27
Slovakia	–€3,992.05
Finland	–€3,222.61
Hungary	–€2,993.62

This is an important analytical result.

For example, Denmark has very high nominal earnings, but its earnings are **below what the regression would predict given its extremely high price level**.

Luxembourg is the opposite: its earnings are dramatically above the level predicted by its PLI.

This provides information that a simple “highest salary” ranking would miss.

12. Affordability Index

The project's central derived measure is the **affordability index**.

The calculation is based on:

$$EarningsToPLI\_i = NetEarnings\_i / PLI\_i$$

This gives a simplified measure of earnings relative to the price-level index.

The result is then normalized so that the EU reference equals 100:

$$AffordabilityIndex\_i = [(Earnings\_i / PLI\_i) / (EU27Earnings / 100)] \times 100$$

Interpretation:

- **100** → EU27 reference level;
- **>100** → above the EU reference;
- **<100** → below the EU reference.

This is the project's primary purchasing-power proxy.

13. Affordability Results — 100% Earnings

The results were:

Rank	Country	Affordability
1	Luxembourg	153.23
2	Austria	120.93
3	Ireland	120.68
4	Netherlands	115.12
5	Denmark	111.59
6	Belgium	110.71
7	Germany	106.30
8	Sweden	106.26
9	Malta	103.22
10	France	103.80
11	Finland	103.41
12	Spain	102.42
13	Cyprus	97.93
14	Italy	93.59
15	Slovenia	93.58
16	Portugal	84.51
17	Lithuania	83.64
18	Poland	81.88
19	Croatia	81.73
20	Czechia	81.28
21	Bulgaria	77.34
22	Romania	75.48
23	Latvia	74.95
24	Estonia	73.55
25	Slovakia	68.37
26	Greece	63.94
27	Hungary	62.13

The most striking result is **Luxembourg**.

Its PLI is among the highest in Europe, but its earnings are sufficiently high that it remains the strongest country under this affordability measure.

This is an important demonstration of why:

> *High prices do not automatically imply low purchasing power.*

What matters is the relationship between earnings and prices.

14. Country Case Studies

14.1 Romania

Romania provides a particularly interesting case for the analysis.

The country has:

- PLI ≈ **65.1**
- Net earnings ≈ **€13,232.74**
- Affordability index ≈ **75.48**

This means Romania is among the least expensive EU countries according to the PLI, but its relatively low earnings more than offset the benefit of low prices in this simplified affordability measure.

Romania therefore illustrates an important analytical distinction:

> *Low cost of living is not necessarily equivalent to high purchasing power.*

A country can be inexpensive relative to the EU while still providing relatively weak purchasing power because local earnings are also substantially lower.

14.2 Bulgaria

Bulgaria represents a similar but slightly different case.

It had:

- the lowest PLI in the analyzed EU sample: **62.5**
- net earnings of approximately **€13,016.65**
- affordability index of approximately **77.34**

Interestingly, Bulgaria has a **positive regression residual** despite its low affordability index.

This means that its earnings are somewhat higher than what the overall earnings-price regression would predict at such a low PLI.

This distinction is useful:

Regression residual

Answers:

> *"Does the country earn more or less than expected given its price level?"*

Affordability index

Answers:

> *"How does the country's earnings-to-price relationship compare with the EU reference?"*

They are therefore related but conceptually different measures.

14.3 Luxembourg

Luxembourg is the clearest high-performing outlier in the analysis.

It had:

- PLI: **131.5**
- Net earnings: **€54,259.93**
- Affordability index: **153.23**
- Regression residual: approximately **+€12,231**

The country therefore combines:

- very high prices;
- exceptionally high earnings;
- the highest affordability index in the project.

This is precisely the kind of result that would be missed by looking only at a cost-of-living ranking.

Luxembourg's result also demonstrates why the analysis should not equate "expensive" with "unaffordable."

14.4 Denmark

Denmark presents the opposite analytical lesson.

It had:

- the highest PLI: **139.7**
- net earnings: **€41,981.24**
- affordability index: **111.59**

Therefore, despite being the most expensive EU country in the project's dataset, Denmark still sits above the EU affordability benchmark.

However, its regression residual was approximately:

–€4,006

This means that, relative to the cross-country earnings-price relationship, Danish earnings were lower than the model would predict given the country's very high price level.

This shows why several complementary metrics are useful.

15. 50% Earnings Scenario

The project also repeated the analysis for workers earning 50% of the average wage.

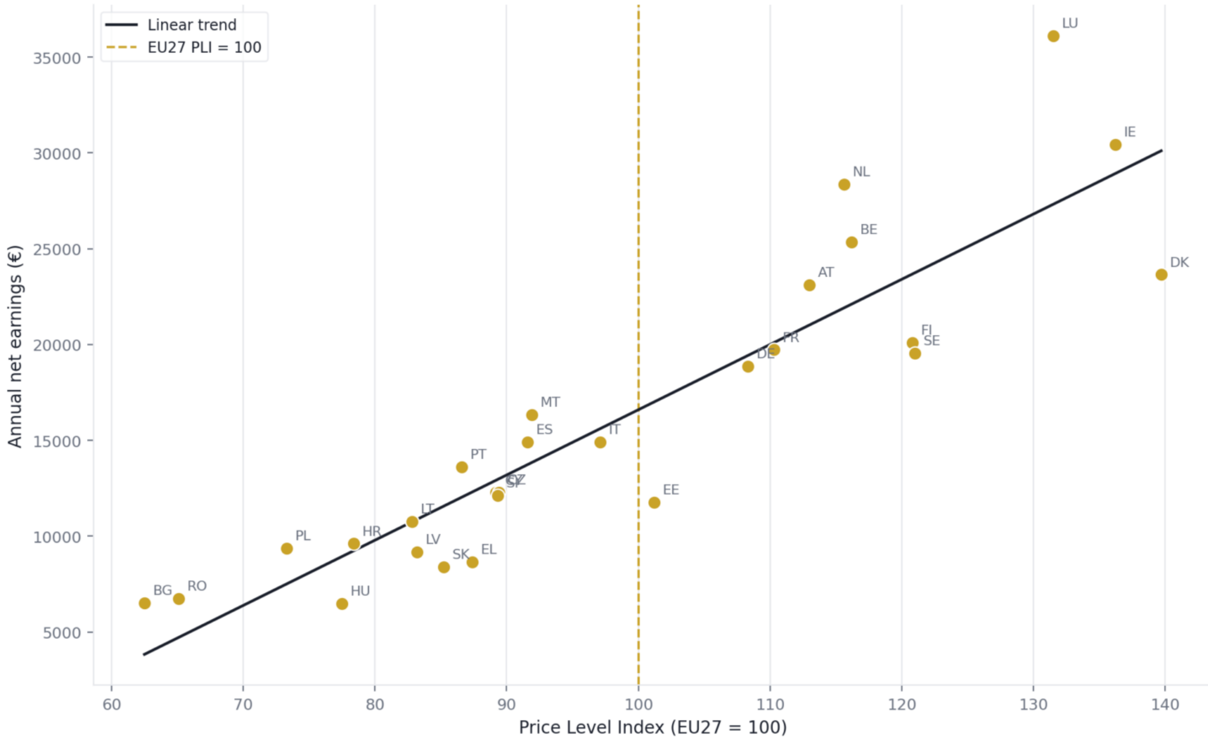

Figure 3. Price levels vs. net earnings 50%

The EU27 reference earnings were:

€16,722.37

The 50%-earnings affordability results showed a lower overall purchasing-power position.

Examples include:

Country	Affordability — 100%	Affordability — 50%
Luxembourg	153.23	164.22
Netherlands	115.12	146.76
Belgium	110.71	130.44
Ireland	120.68	133.62
Austria	120.93	122.34
Romania	75.48	61.93
Bulgaria	77.34	62.27
Hungary	62.13	50.03

Country	Affordability — 100%	Affordability — 50%
Greece	63.94	59.19
Slovakia	68.37	58.88

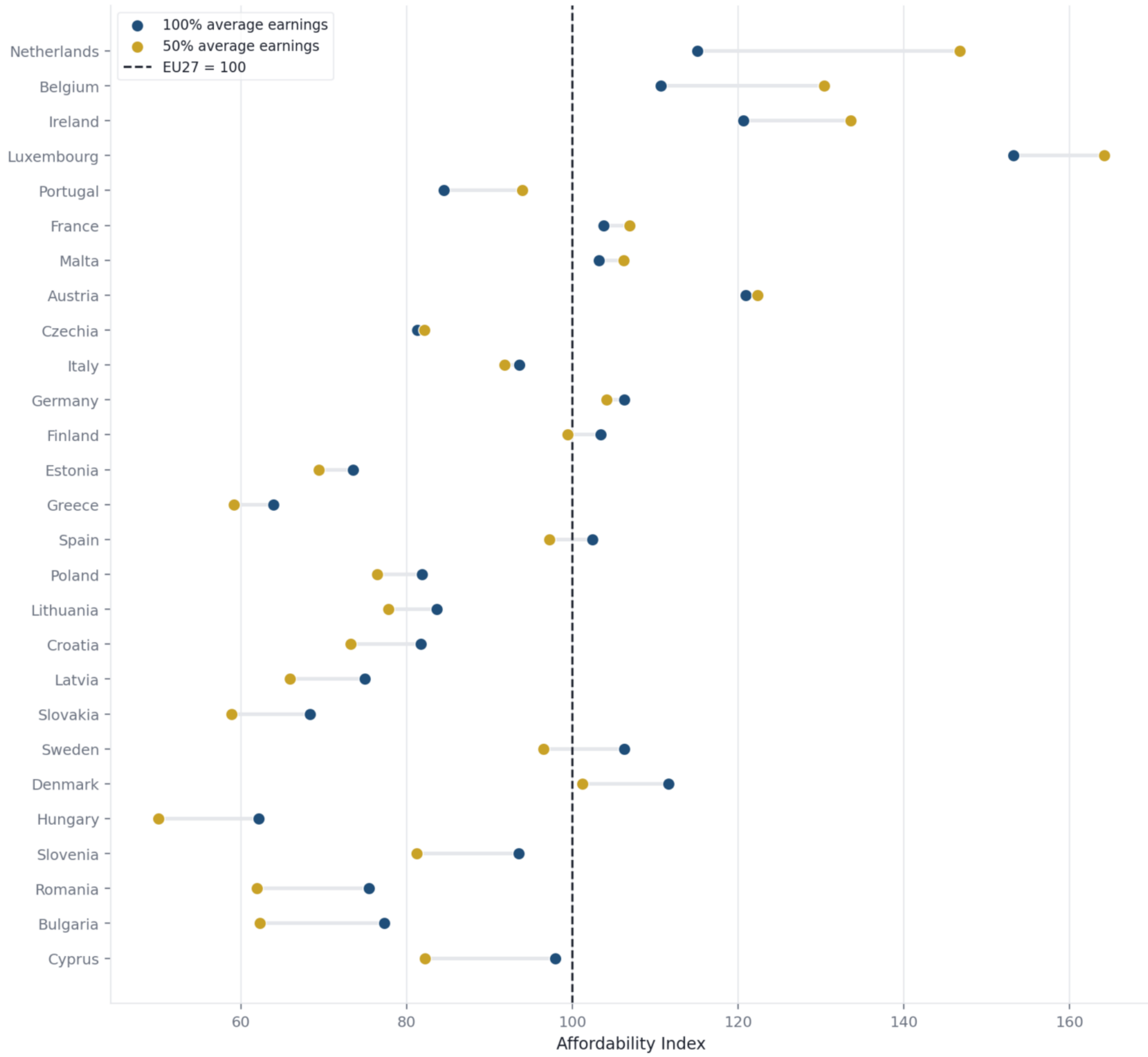

Figure 4. Affordability 100% vs 50%.

An important observation is that the relationship is **not simply a uniform 50% reduction** in the final index because the 50%-earnings scenario is based on a separate Eurostat earnings dataset and corresponding country values.

The scenario therefore provides a useful sensitivity analysis of how the affordability picture changes under a lower-income worker profile.

16. Change in Affordability

The project calculated:

$$AffordabilityChange = Affordability\_50 - Affordability\_100$$

The largest positive changes included:

Country	Change
Netherlands	+31.64
Belgium	+19.73
Ireland	+12.94
Luxembourg	+11.00
Portugal	+9.46

The largest negative changes included:

Country	Change
Cyprus	–15.71
Bulgaria	–15.07
Romania	–13.56
Slovenia	–12.37
Hungary	–12.10

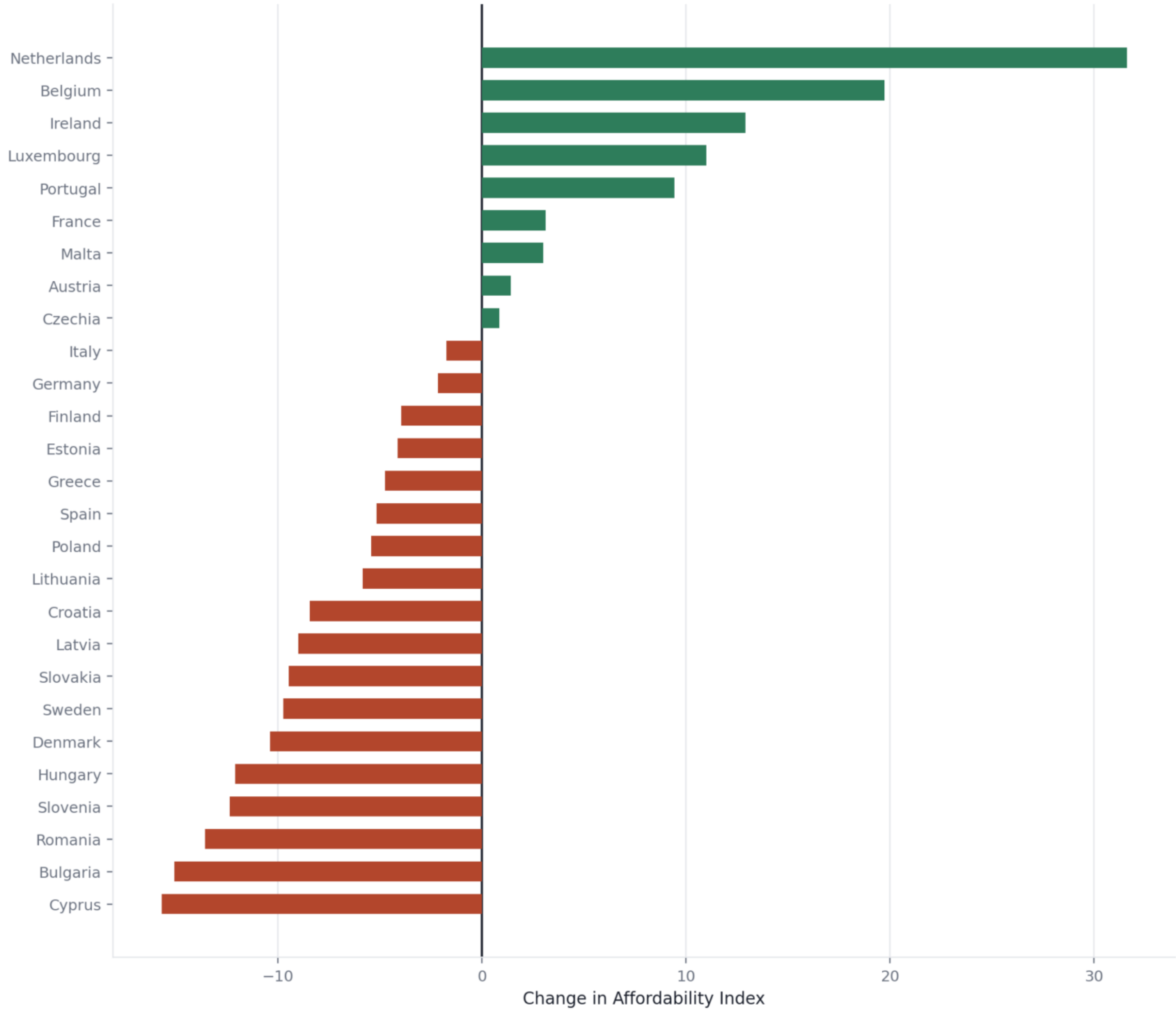

Figure 5. Affordability change.

This comparison highlights that the relationship between income scenario and affordability is not identical across countries.

17. Why the Regression and Affordability Index Should Both Be Used

One of the strongest aspects of the project is that it does not rely on one metric.

The analysis provides three complementary perspectives.

1. Net earnings

Measures:

> *How much money does the worker earn?*

2. Regression residual

Measures:

> *Does the country's earnings level exceed or fall below what would be expected from its price level?*

3. Affordability index

Measures:

> *How strong is the country's earnings-to-price relationship relative to the EU benchmark?*

This produces a more nuanced analysis than a simple ranking.

18. Power BI Dashboard

The analytical results were translated into an interactive Power BI dashboard.

The dashboard includes:

KPI cards

Providing high-level indicators such as:

- average net earnings;
- affordability measures;
- reference values;
- other headline statistics.

Affordability visualization

Shows the relative affordability position of EU countries.

Scatter plot

Visualizes:

> *Price Level Index vs. Net Earnings*

with:

- country labels;
- regression/trend relationship;
- EU reference;
- earnings information.

Affordability change chart

Compares:

> *100% earnings vs. 50% earnings*

and uses conditional formatting to distinguish positive and negative changes.

Country slicer

Allows users to select one or multiple countries.

This makes the dashboard useful both for:

- high-level comparison;
- individual country exploration.

19. Streamlit Application

In addition to Power BI, the project was developed into an interactive **Streamlit application**.

The Streamlit application provides a Python-native interactive interface and complements the Power BI dashboard.

The application includes:

- headline KPIs;
- affordability analysis;
- earnings and PLI comparison;
- interactive scatter analysis;
- country-level detail;
- 100% vs. 50% earnings comparison.

The presentation layer was subsequently refactored to improve:

- spacing;
- visual hierarchy;
- typography;
- consistency;
- section organization;
- overall professional appearance.

The country-detail section also uses the full country name rather than displaying only Eurostat geographic codes.

This gives the project two distinct presentation environments:

- **Power BI** → business intelligence / dashboard presentation
- **Streamlit** → interactive Python analytics application

20. Technical Implementation

The project uses:

- Python
- pandas
- NumPy
- matplotlib
- Streamlit
- Power BI
- Git/GitHub

The Python pipeline separates:

```
Raw data
  ↓
Data loading
  ↓
Validation
  ↓
Filtering
  ↓
Dataset merging
  ↓
Derived metrics
  ↓
Statistical analysis
  ↓
Visualization
  ↓
Processed dataset
  ↓
Power BI / Streamlit
```

The processed analytical dataset contains variables such as:

```
geo
Geopolitical entity (reporting)
pli
net_earnings_eur
predicted_earnings
earnings_residual
earnings_index
earnings_to_pli
affordability_index
net_earnings_50_eur
predicted_earnings_50
earnings_residual_50
earnings_index_50
earnings_to_pli_50
affordability_index_50
affordability_change
```

This structure makes the final dataset suitable for both statistical analysis and BI visualization.

21. Reproducibility

An important design goal was reproducibility.

Instead of performing all transformations manually inside the notebook, the project moved reusable logic into src/.

For example, the regression function is generalized:

```
def fit_linear_regression(x, y):
```

rather than being hard-coded specifically for PLI and earnings.

Likewise, the affordability calculation accepts configurable columns and a suffix:

```
def compute_affordability_index(
    df,
    pli_col,
    earnings_col,
    eu27_earnings,
    suffix=""
):
```

This allows the same function to calculate:

```
affordability_index
```

and:

```
affordability_index_50
```

without duplicating analytical logic.

That makes the pipeline more maintainable and easier to extend to other scenarios.

22. Limitations

The project has several important limitations.

25.1 Affordability is a proxy

The affordability index is **not an official Eurostat purchasing-power measure**.

It is a project-specific analytical indicator based on:

$$NetEarnings / PLI$$

It should therefore be interpreted as a **simplified purchasing-power proxy**.

25.2 PLI is not a complete cost-of-living measure

The PLI covers household final consumption expenditure, but the cost experienced by an individual depends on their specific consumption basket.

For example:

- housing;
- food;
- transport;
- healthcare;
- education;
- energy

can differ considerably in importance between individuals.

Eurostat itself publishes more detailed PLI categories because overall household consumption is only one level of analysis.

25.3 Gross vs. net earnings distinction

The project uses **net earnings**, which is appropriate for an affordability-oriented analysis because net income is closer to the amount actually available to households.

However, differences in:

- taxation;
- social contributions;
- benefits;
- household structure

can influence the comparison.

25.4 Worker profile

The analysis focuses on a specific worker profile:

> *Single person without children earning 100% of average earnings.*

Therefore, the results should not automatically be generalized to:

- families;
- households with children;
- part-time workers;
- retirees;
- unemployed people;
- high-income workers;
- minimum-wage workers.

25.5 Cross-sectional analysis

The main analysis focuses on **2025**.

Therefore, the regression identifies a cross-country relationship for one period rather than a time-series relationship.

A future version could examine:

```
2015 → 2016 → ... → 2025
```

to investigate how affordability has evolved over time.

25.6 Correlation does not imply causation

The correlation of **0.937** is strong, but it does not demonstrate that higher prices cause higher wages.

The relationship likely reflects broader economic differences between countries.

The regression should therefore be interpreted as **descriptive**, not causal.

23. Key Findings

The project produces several major findings.

Finding 1 — Price levels differ dramatically across Europe

The 2025 household-consumption PLI ranged from approximately **62.5 in Bulgaria** to **139.7 in Denmark**. Eurostat independently reports the same broad range.

Finding 2 — High prices tend to coexist with high earnings

The correlation between PLI and net earnings was **$r = 0.937$** with **$R^2 = 0.877$**. This represents a very strong descriptive relationship in the 27-country sample.

Finding 3 — High salaries do not automatically mean poor affordability

Luxembourg demonstrates this particularly clearly. It has very high consumer prices, but its exceptionally high net earnings result in the highest affordability index.

Finding 4 — Low prices do not automatically mean strong purchasing power

Romania and Bulgaria have among the lowest price levels in the EU, but their lower earnings result in affordability indices below the EU benchmark.

Finding 5 — Denmark illustrates the importance of relative analysis

Denmark has the highest PLI, but still has an affordability index above 100. Its position would look considerably worse if only price levels were considered.

Finding 6 — Income scenarios matter

The 50%-earnings scenario changes the relative affordability picture significantly for several countries. This demonstrates why a single “average salary” metric is insufficient for understanding household economic conditions.

Finding 7 — The regression identifies important outliers

Luxembourg's earnings are dramatically above the regression expectation, while Estonia and Greece are among the countries with the largest negative residuals. This provides a useful second layer of analysis beyond simple rankings.

24. Overall Interpretation

The central conclusion of the project is:

> *Cost of living should not be evaluated independently from income.*

A country can be:

- expensive but highly affordable;
- inexpensive but relatively unaffordable;
- high-income but expensive;
- low-income but inexpensive.

The economically meaningful comparison is therefore not simply:

> *"Where are prices lowest?"*

or:

> *"Where are salaries highest?"*

but rather:

> *"How does the income available to a representative worker compare with the price level they face?"*

The project attempts to answer this question through the affordability index.

25. Project Value

From a data analytics perspective, the project demonstrates a complete workflow:

Data acquisition

Working with real-world Eurostat datasets.

Data cleaning

Handling metadata-heavy statistical exports and selecting relevant observations.

Data validation

Checking missing values, duplicates, geography and reference observations.

Data transformation

Merging earnings and PLI datasets.

Statistical analysis

Using correlation, linear regression, R^2, residual analysis.

Feature engineering

Creating earnings index, earnings-to-PLI, affordability index, scenario comparison.

Data visualization

Creating ranked charts, scatter plots, regression visualizations, dumbbell/change charts.

Business intelligence

Building an interactive Power BI dashboard.

Application development

Building an interactive Streamlit application.

Software engineering

Organizing reusable functions into a src package and maintaining the project through Git/GitHub.

This makes the project considerably broader than a simple exploratory notebook.

26. Future Improvements

Several extensions would make the analysis significantly stronger.

Historical analysis

Instead of only 2025:

```
2015–2025
```

could be analyzed to determine whether affordability has improved or deteriorated.

More detailed consumption categories

Instead of overall household consumption, the project could examine:

- food;
- housing;
- transport;
- energy;
- restaurants;
- recreation;
- healthcare;
- education.

This would allow the creation of category-specific affordability measures.

Eurostat's PPP data provides detailed categories for this purpose.

Household scenarios

Additional profiles could include:

- single person;
- couple;
- couple with children;
- minimum-wage worker;
- median worker;
- high-income worker.

Housing-adjusted affordability

Housing is one of the largest household expenditure categories and varies significantly between countries. Eurostat's 2025 publication shows particularly large differences in housing-related price levels.

A housing-adjusted index could therefore provide a more realistic measure of disposable purchasing power.

More sophisticated statistical models

The linear regression could eventually be extended using:

- multiple regression;
- robust regression;
- clustering;
- PCA;
- panel-data models.

Additional explanatory variables could include:

- GDP per capita;
- productivity;
- unemployment;
- taxation;
- housing costs;
- social expenditure.

27. Final Conclusion

This project demonstrates that comparing European living standards requires more than comparing salaries or consumer prices individually.

The 2025 data reveals a strong relationship between national price levels and net earnings, but the relationship is far from uniform. Countries such as Luxembourg demonstrate that exceptionally high earnings can compensate for high prices, while Romania and Bulgaria demonstrate that very low prices do not necessarily translate into high purchasing power.

The project's affordability index provides a practical way to combine these two dimensions into a single comparative measure. The regression and residual analysis then adds another layer by identifying countries whose earnings are unusually high or low relative to their price level.

The overall analysis therefore moves through three increasingly meaningful questions:

> *How much do things cost? → PLI*
>
> *How much do people earn? → Net earnings*
>
> *How much purchasing power does that income imply? → Affordability index*

The combination of **Python analytics, statistical modelling, Power BI and Streamlit** turns the project into a complete end-to-end data analytics portfolio piece, demonstrating not only the ability to analyze a dataset but also to transform raw statistical data into reusable analytical logic, interactive dashboards and an interpretable economic narrative.

Data references

The primary data sources are Eurostat's **Annual net earnings (earn_nt_net)** dataset and its **Purchasing Power Parities / Price Level Indices (prc_ppp_ind_1)** data.

Eurostat — Annual net earnings dataset

Eurostat — Purchasing Power Parities